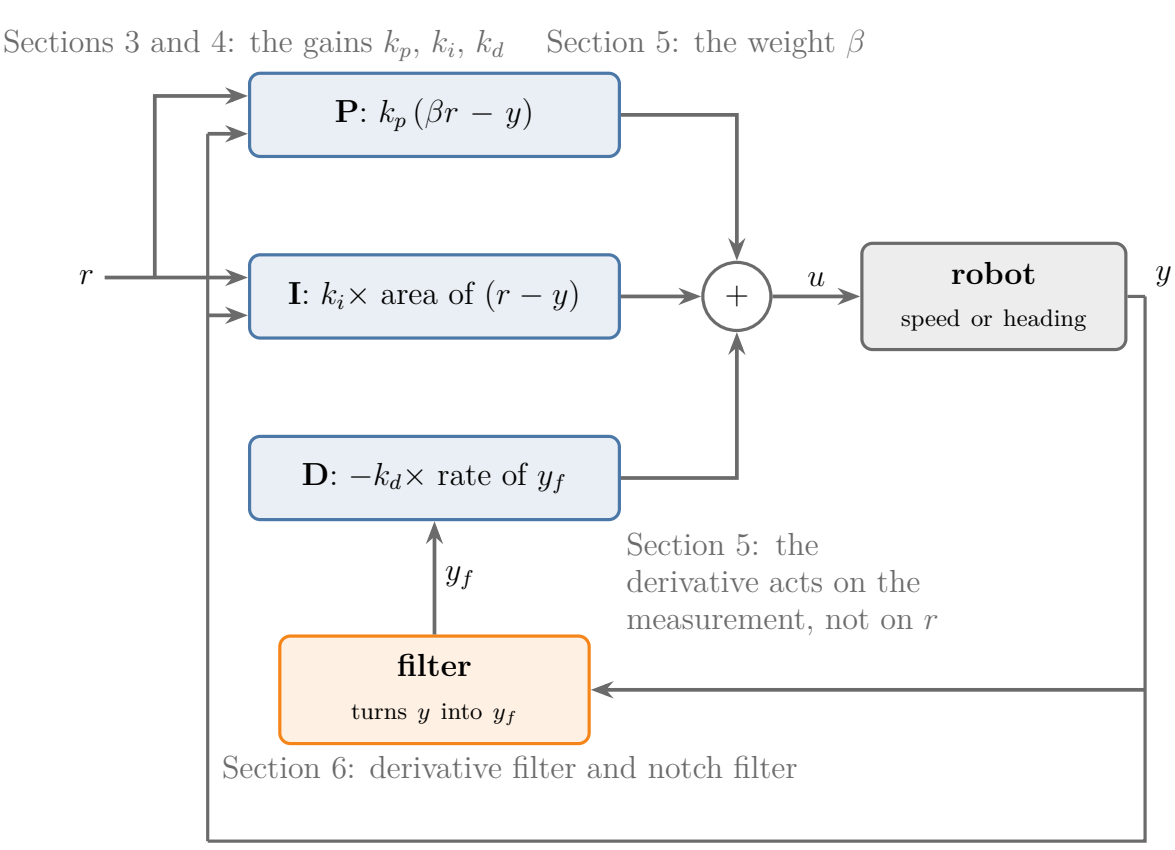

Sections 3 and 4: the gains k_p, k_i, k_d
Section 5: the weight β
P: $k_p\,(\beta r - y)$
r
I: $k_i\times$ area of $(r - y)$
+
u
robot
speed or heading
y
D: $-k_d\times$ rate of y_f
Section 5: the derivative acts on the measurement, not on r
y_f
filter
turns y into y_f
Section 6: derivative filter and notch filter